

Deep Learning Model Architectures for Medicinal Plant Identification and Classification: A Comparative Evaluation

1st Yamini Singh

CSE Department, Amity School of
Engineering & Technology
Amity University Uttar Pradesh
Noida, India
yamini1599singh@gmail.com

2nd Sanjay Kumar Dubey

CSE Department, Amity School of
Engineering & Technology
Amity University Uttar Pradesh
Noida, India
skdubey1@amity.edu

Abstract— It is very important to identify medicinal plants correctly in pharmacognosy, conservation and natural product research. Convolutional Neural Networks (CNNs) and other deep learning (DL) models have proven quite promising in automating such a task. Recently, more popularity of Convolutional Neural Networks (CNNs) in image classification-based medicinal plants classification is witnessed because they can extract and study intricate visual characteristics. People are talking about the compromises involved in achieving practical deployment, accurate results and high machine efficiency. Models that are more practical can usually be used with less effort, yet ensemble models demonstrate better performance but need more resources. The purpose of the study is to look at and compare the benefits and drawbacks of each DL model individually, transfer learning and using groups of models as ensembles. When performing benchmark tests, accuracy, how much time is required for training and the model structure are all considered. The highest result is from an ensemble group named EDL-AMLI that consists of MobileNetV2, InceptionV3 and ResNet50, producing 99.66% accuracy in testing and 99.9% accuracy in cross-validation. Nonetheless, the transfer learning utilizing InceptionV3 with SVM had best standalone accuracy (99.81%) and balanced training cost. In this comparative study, an overall performance scale has been taken into consideration to set the framework of a future study, and directions to be taken into consideration have also been provided. To enhance the interpretability, another route in this study is the integration of Explainable AI (XAI). Future opportunities, the review notes, include ensuring explainable AI, but at the same time it puts great weaknesses in terms of the current situation, where it is quite difficult to generalize and the amount of available data is just insufficient. This research helps in picking the proper deep learning models for cases related to medicinal plants.

Keywords— *Deep Learning, Transfer Learning, Ensemble Learning, Convolutional Neural Networks (CNN), VGG16, VGG19, ResNet50, EfficientNet, InceptionV3, DenseNet121, MobileNetV2, Support Vector Machine (SVM).*

I. INTRODUCTION

Plants have been an important source of medicine from ancient times, the plant kingdom was considered as a treasure house of many useful medicines. Even today, the World

Health Organization estimates that up to 80 percent of people still rely mainly on traditional remedies such as herbs for their medicines Sathasivam and Muthuselvam [1]. There are multiple plant types around us, out of those some are used for medicinal properties and others can be poisonous as well. Therefore, there is a need for such system which can help to recognize the medicinal plants in a better and easy way. Technology using deep learning has grown to become an essential support in areas like researching herbal medicine, saving wildlife and health applications. Convolutional Neural Networks (CNNs) have cut down false identifications by a considerable margin employing especially their capacity to enhance high-level features on multifaceted plant images. Stella et al. [2] focused on Xception architecture as a tool of recognizing medicinal plants and accentuated the role of deep models on accurate classification.

In a similar manner, Nguyen Quoc and Hoang [3] demonstrated that CNNs can be trained to classify up to 200 species using transfer learning and the CNN accuracy depended on dataset size and background noise. Several investigation have focused on many deep learning techniques, ranging from single CNN networks to combining various models. Dileep and Pournami [4] applied SVM classifiers on AyurLeaf dataset and pointed to accuracy improvement when image augmentation was used and Islam et al. [5] suggested mobile compatible classifier in order to identify leaf quickly.

Prasannan et al. [6] stated the importance of having uniform data sets and more versatile models that could support the variability of plants in real conditions. In more recent work, including Kar and Kar [7], it has been shown how classifiers work using attention mechanisms and vision transformers with transfer learning work to improve classification performance.

The more cutting-edge technologies used in contemporary medicinal plant recognition are EfficientNet and XGBoost for modular tasks S. S. et al.[8] and Vision Transformer (ViT) structure structures in fine-baud prognosis of a complicated commodity, on location of better congruence and scale in herbal and intellection aspects T. R. et al. [9].

II. LITERATURE REVIEW

A. Individual Model Based Studies

Recent advances in ML and DL have made it easier to identify medicinal leaves and benefit agriculture, the conservation of biodiversity and environmental supervision. DL and ML have made data collection and analysis quicker and more accurate, the weak points of traditional manual identification are decreased.

Bandla et al. [10] in the study, sampled and identified the main features of medicinal leaves using the well-known models MobileNetV2 and AlexNet. The efficiency and ability to handle large data make it easy to examine large sets in a number of situations. According to the read paper that used picture cleaning, segmenting of images and feature extraction can aid in learning and improving accuracy in prediction. MobileNetV2 was unique in that it performed the best when large sets were taken. In Acasamoso [11] strides to distinguish three groupings of medicinal plants, they managed to attain training and validation accuracies of 99.22 percent and 84.38 percent, respectively. This study demonstrates that deep learning is promising in classification tasks that are not very complex, seeing as future researchers can recommend additional plant families and geo-tagging in order to increase their scope. V. H. A et al. [12] highlighted that CNN, InceptionV3, and ResNet50 were adopted in the detection of tulsi, peepal, and neem, with InceptionV3 providing the best classification rate of 100 percent and quite low loss of 0.0199 as compared to ResNet50. Hussain et al. [13] have used specially designed Extended CNN models that have performed well with 82.29 and 76.93 percent of accuracy and precision, respectively, proving their efficacy even when the image datasets are limited and imperfect, and needing only 31.42 seconds of training time. An example of a study is by Prathima et al. [14], who studied CNN, and MobileNet on medicinal plant classification and concluded that MobileNet resulted in fewer overfitting and better generalization compared to CNN, and outperformed in training and validation. Salian et al. [15] suggested MPI-net that utilises VGG16 which was trained on 3,000 images of 10 species and received 99.4 percent accuracy, 99.6 percent precision which could be applied to an Android application and used in the classroom to teach about the species. Aquiro et al. [16] trained ResNet50 model to classify 20 herbal plants with the accuracy of 95.17% and provided a GUI to display the plant name and potential health applications. Paulson and Ravishankar [17] contrasted CNN, VGG16 and VGG19 and they found VGG16 and VGG19 superior with the accuracies of 97.8 percent and 97.6 percent, respectively. Sonkar et al. [18] presented a CNN model that could mark fungal problems in leaf images with 98.87 per cent precision and was trying to enhance it by using additional data and an easy-to-understand interface. Kikome et al. [19] proposed an attention-based model consisting of CNNs to achieve 98 percent accuracy in the training set, validation set, and test set and also works in simple and complex structures of a leaf. Bhutada et al. [20] stressed that CNNs would be faster and more accurate in identifying plants and support cost-efficient measures and retain the old knowledge. CNN model developed by Saunshi et al. [21] performed remarkably recognizing the plants and their

medicinal properties, with a 98 percent accuracy and the work made a landmark in the context of the practice of alternative medicine. Begam et al. [22] scanned using the VGG16 model and detecting rare medicinal plants, with 98 percent accuracy, the performance was improved according to training time. The Enhanced CNN, ResNet, and YOLOv5-X were used by Sangeetha et al. [23] in the disease detection of crops and proved to be rather accurate in classification and confidence based predictions with heavy preprocessing. In very uncontrolled conditions, Pushpanathan et al. [24] tested models on 34,200 images of 12 Asian herbs, and found VGG16 and VGG19 did best with the variation in lighting conditions and angle. According to the results of Chandra et al. [25], the InceptionV3 is the most accurate classifier, with 98.20%, and it outperforms Random Forest and SVM classifier to be used in LTECP-based classification system, and the use of deep learning has many advantages that can be enhanced with an optimized feature extraction.

B. Ensemble Model based study

Pandey et al. [26] pointed out a significant way is to create an ensemble method with deep learning models such as VGG16, ResNet50, and ResNet152, because it helps the model to predict more accurately. The strategy increases certainty by merging predictions from several models and picking the prediction made most often. Results with VGG16 (98.42%), ResNet50 (99.11%) and ResNet152 (97.02%) are much better than traditional, handcrafted feature-based methods. Sourov et al. [27] invented a different automatic system that is based on artificial neural networks (ANN) and convolutional neural networks (CNN) to distinguish and classify medicinal plant species. Working with 5000 images of 10 medicinal plant leaves, the system achieves a remarkable 99 percent of accuracy output. In the experiments carried out by Custodio et al. [28] with seven settings of the convolutional neural networks and the five transfer learning models, ResNet50, ResNet101, ResNet152, VGG16 and VGG19 were all used. The best precision came out of bringing all five models together. From the study, increasing the dataset with various plant structures like stems, barks and flowers could boost the model's performance. Shafik et al. [29] in the study noticed that it produces two version of ensemble models for plant disease detection by combining nine pretrained CNNs. It implies that the results on accuracy, reliability and generalization are superior when using these networks over single CNNs. Logistic regression is fitted into the classification system to make the decision process even better. Ghosh et al. [30] noticed that the test results reveal that the suggested model does better than others in identifying medicinal plant leaves. The hybrid model reached 95% accuracy which is higher than what was seen with the separate pre-trained models. PCA makes the model better by removing features that are not important which makes it able to generalize more effectively. A research by Pushpa et al. [31] presented three new hybrid models that join the benefits of various deep convolutional neural networks to face these challenges. This model with the use of MobileNetV2 and Squeeze-and-Excitation (SE) layers helped to raise the accuracy score, reaching 94.24%. Emphasizing feature enhancement using hybrid architectures was a notable part of the research and model 3 showed the best results because of

better feature scaling and recalibration, even when dealing with a relatively small data set. Naeem et al. [32] propose a machine learning classifier for leaf patterns comparison of different medicinal plants. The multi-layer perceptron performed best, of all these models, having an accuracy of 99.01%. Because medicinal plants can cure many diseases, fast and accurate identification of these plants is very important. The research by Sachar and Kumar [39] uses deep learning which is a branch of Artificial Intelligence and implements an ensemble approach to identify medicinal plants from leaf pictures. The Ensemble Deep Learning-Automatic Medicinal Leaf Identification (EDL-AMLI) model which averages many networks, performed better than the standalone pre-trained models.

C. Transfer Learning Approach based study:

Ayumi et al. [33] adopted VGG16, VGG19 and MobileNetV2 models as models of their transfer learning to identify leaves of medicinal plants. Of the models tested, the value of validation accuracy (96.02%) and testing accuracy (81.82%) upon fine-tuning was the highest when using MobileNetV2. It describes that in order to utilize data we have to adapt pre-trained model to adapt to its format. To preserve and spread local knowledge about ethnobotany, Azmeraw et al. [34] have recently used artificial intelligence techniques to predict what Ethiopia's medicinal plants can treat. SVM showed the top validation correctness at 99.34% and the model InceptionV3 achieved this result by running an average of SoftMax and SVM. Senanayake et al. [35] looked at the performance of transfer learning models. VGG 16 got 90% accuracy in detecting medicinal plant disease and MobileNet v2 got 99.5% accuracy in categorization of medicinal plants. The use of a lightweight model called MobileNet V2, helps in engineering gadgets suitable for this job. Consequently, suggested future research directions are to increase the number of classes and sample data and to build an Android app for real-time identification and sorting of medicinal plants. Sahib et al. [36] applied six advanced pretrained models and greatly outperformed earlier methods that worked with the same dataset. Bit_s-r50x1 was found to have an even higher accuracy (99.62%) in classification than the existing best method (DenseNet121 with 87.4% accuracy) by a noticeable 5.2%. The experiments confirm how much less time is needed to train the model with the help of transfer learning, while it maintains top performance. When these models are added to mobile or diagnostic systems, they may help biodiversity conservation, herbal medicine studies and make healthcare more accessible for the public. A study by Habiba et al. [37] tells since CNNs can learn different levels of features, they are often used in image categorisation. Since VGG16 is not as complicated, it can work smoothly with short datasets that lack many samples, even though it is larger than some other models. Frequently, classes that look the same are classified incorrectly, particularly for small datasets, which shows why keeping a balance between how detailed the model is, how big the dataset is, and how diverse the data is important.

III. REVIEW METHODOLOGY

The purpose of the literature review is to find, review, select and combine strong research studies for the study. Majorly, a literature review wants to give you an accurate and detailed view of the main research problem using a planned and systematized technique. This paper gathers information from a group of judiciously picked journal articles and research papers. During the first planning phase, the need for the review, proper questions and the approach are all set up. The identification of related studies happens during review based on preset standards, research quality checklists are used to filter them, and data is collected using set procedures. By the end, the collected data is combined, the findings are put into a review and plans for sharing and reporting are arranged.

A. Individual Model

There are many deep learning models which have been working for medicinal plant identification and classification, each of them offers some specific strengths in terms of accuracy, efficiency, and computational cost. Here, the commonly found individual models were reviewed with reference to the underlying theories, the observed performance of the same, and their implications in practice. Pushpanathan et al. [24] showed that VGG16 and VGG19 were ranked high in the uncontrolled conditions, including nature backgrounds and non-uniform lighting. Although such-like architecture has its merits, as per the case presented by Pushpanathan et al. [24], ResNet50 did not perform well in high lighting and viewpoint variability scenes. According to Pushpanathan et al. [24], EfficientNet follows suit in that it is applicable in low-power scenarios but increasingly lacked robustness in the field. Chandra et al. [25] have presented a hybrid pipeline that consisted of InceptionV3 to extract features, chi-square to identify features, and SVM, and Random Forest (RF) to perform the classification. According to Prathima et al. [14] MobileNet was determined to be better than standard CNN in terms of generalization and the efficiency of deployment. Hussain et al. [13] have made a shallow and rapid-learning CNN to address raw medicinal plant images and not require profound preprocessing. Dalvi et al. [38] proposed specific CNN, adapted to the task of recognizing plants in dermatology, which takes as an input several attributes: shape, color, and texture of the leaf. Our classification of the models can be as follows: standard CNNs (e.g. VGG16, VGG19), residual networks (e.g. ResNet50), efficient models (e.g. EfficientNet B0B7), lightweight models (e.g. MobileNet), hybrid models (e.g. InceptionV3 with SVM/Random Forest) and custom-hardcoded CNNs adapting to medicinal settings. The models themselves were compared on which were the most accurate, trained the quickest, were easy to deploy, and easy to adapt to the real world (e.g. circumstance of light, clutter). The mathematical ideas supporting these models were considered to draw out their calculation behaviour and learning performance. For example, if ResNet50 use residual learning with simple identity mapping as:

$$y = F(x, \{W_i\}) + x \quad (1)$$

in which $F(x, \{W_i\})$ is the residual mapping to be learned that makes deep networks stable, avoiding vanishing

gradients. The compound scaling scheme of EfficientNets squares off network's depth, width, and resolution:

$$\begin{aligned} \text{depth} &= \alpha^\phi, \text{width} = \beta^\phi, \text{resolution} = \gamma^\phi \\ \alpha \cdot \beta^2 \cdot \gamma^2 &\approx 2 \end{aligned} \quad (2)$$

MobileNet is a type of CNN that will use depthwise separable convolutions to cut down cost of computation, which is computed as:

$$\frac{\text{Depthwise Separable Cost}}{\text{Standard Conv Cost}} \approx \frac{1}{K^2+1} \quad (3)$$

where the above equation is of $K \times K$ kernels. The results of performance of the chosen studies were also discussed. As an example, MobileNetV2 fine-tuned by Ayumi et al. [33] attained validation and testing accuracy of 96.02 percent and 81.82 percent respectively, indicating its high test/train generalization with small data. Equally, a combination of InceptionV3 and SVM classifier achieved a validation accuracy of 99.34% in study of Azmeraw et al. [34] and shows the advantage of the hybrid model of feature extractors and classifier. The sizes of available data sets in the reviewed studies varied significantly and ranged between small-scale 80:20: testing splits of e.g., 1500 images to several thousand images with more than ten classes. Other preprocessing techniques were also recorded like segmentation is used to reduce background noise as well as chi-square feature selection to interpret classifiers easily. This approach allows to inspect, not only the performance results, against which models can be compared, but also to inspect the theoretical soundness, real-world applicability, and architectural scalability. Such features are of great importance in relation to the uses in biodiversity preservation, ethnobotany studies, and healthcare diagnostic transient.

B. Ensemble Learning Approach:

Ensemble learning the strategies that allow a combination of models usually with varied architectures to make more resounding and stable predictions than models taken separately. The method is especially useful with tasks having a high variance of activities such as the identification of medicinal plant species using images. In all the reviewed studies, ensemble techniques always performed better than standalone CNN and transfer learning and detected more and complementary patterns of diverse inputs. Model fusion is powerful as it resulted in 99.11 percent accuracy in the ensemble of VGG16, ResNet50, and ResNet152 in the studies by Pandey et al. [26]. It was demonstrated by Sourov et al. [27] that averaging the outputs of VGG and ResNet models optimized by a classical fine-tuning procedure achieved the classification accuracy of 99% when classifying ten plant species. Custodio et al. [28] applied hybrid model of ResNet and VGG, in which they averaged the ensemble successfully and achieved better performance than models applied separately.

Let $h_1(x), h_2(x) \dots h_n(x)$ be predictions from n base classifiers on an input x . The final prediction denoted by $H(x)$ from an ensemble model can be computed using one of the below mentioned methods:

a) Majority Voting:

$$H(x) = \arg \max_{y \in Y} \sum_{i=1}^n \mathbb{I}[h_i(x) = y] \quad (4)$$

where $\mathbb{I}$ is the indicator function (1 if true, 0 otherwise) and each classifier votes for a class label, and the class with the highest votes is selected.

b) Weighted Voting:

$$H = \arg \max_{y \in Y} \sum_{i=1}^n w_i \cdot \mathbb{I}[h_i(x) = y] \quad (5)$$

Weights w_i are based on model performance (e.g., validation accuracy).

c) Stacking (Meta-Learning):

A meta-model g is trained on the outputs of base classifiers:

$$H(x) = g(h_1(x), h_2(x) \dots h_n(x)) \quad (6)$$

The above-mentioned meta-model g can be a logistic regression, neural network, or Support Vector Machine.

d) Early Fusion:

$$f_{\text{fusion}}(x) = f_1(x) \oplus f_2(x) \oplus \dots \oplus f_n(x) \quad (7)$$

Instead of combining predictions, early fusion merges intermediate features (e.g., CNN feature vectors):

where $\oplus$ is concatenation or averaging of feature vectors.

The feature extraction in PCA-enhanced VGG was introduced by Ghosh et al. [30] to optimize the results by selecting features prior to the ensemble voting, and a squeeze-excitation CNN (SE-CNN) illustrated by Pushpa et al. [31] used multilateral grouped feature extractors to raise the classification accuracy. According to Shafik et al. [29], applying nine CNN models resulted in a considerable increase in accuracy and an increase in memory and computing expenses. Aarthi et al. [34], within the frame of the transfer learning, were able to see that the generalization of the leaf datasets was improved by combining MobileNetV2, VGG16, and DenseNet201 models together in an ensemble. Mohan et al. [35] concluded that an ensemble of ResNet152V2 and VGG19 by weighted combination of prediction raised the reliability of the model. Bandi and Geetha [36] applied ensemble fusion to a trained DenseNet and had a greater level of precision in the classification of medicinal plants. The ensemble of MobileNetV2, InceptionV3 and ResNet50 led Sachar and Kumar [39] to attain an accuracy of 99.66%, thus showing that diversity in the ensemble helps in improving the classification performance. Along with the above advantages come the limitations of having to train more complex ensembles and not having the corresponding improvement after a specific number of models is used. To sum up, ensemble learning, particularly combined with transfer learning, can be very beneficial in medicinal plant identification and high accuracy and minimization of variance coupled with robustness of image-based classification systems may be achieved.

C. Transfer Learning: It is called Theory and Applications in Plant Classification.

1) *Transfer learning Information:* Transfer learning (TL) is a machine learning concept in which a model learned to

accomplish a large-scale source task is reused on a different target task. TL is an approach that modifies features learned during training without re-training the models, often features learnt in convolutional neural networks (CNNs) are pre-trained on large datasets, on the task of interest, such as identifying medicinal plants, which have relatively small quantities of available training data. Let $D_s = \{X_s, Y_s\}$ represents the source and $D_T = \{X_T, Y_T\}$ represents the target domains. Transfer Learning look to strengthen the learning of a predictive function $f_T(\cdot)$ in D_T using knowledge which is from derived $f_s(\cdot)$ in D_s , provided $D_s \neq D_T$ and $T_s \neq T_T$. Using Transfer Learning becomes useful when we have few and mixed data for plant classification.

2) The strategies for Transfer Learning:

a) *Feature Extraction*: In feature extraction strategy, the convolutional base of a pre-trained model is frozen and used to extract general features.

b) *Fine-Tuning*: Generally, upper layers in the pre-trained model are unfrozen, along with the new classifier and they are both retrained. This enables an adjustment to the domain specifics.

c) *Hybrid Transfer Learning*: In the concept of Hybrid Transfer Learning deep feature extraction is combined with machine learning classifiers such as Support Vector Machine, Random Forest, or XGBoost for better performance in certain scenarios.

Transfer Learning makes it possible to adjust models to tasks with few domain-specific data with the help of the models trained on a much larger dataset, e.g. medicinal plant classification. It is practical to re-use generalized features (e.g., shape, texture) on different settings and thus well suited when presented with different and small plant datasets. Its greatest advantages are fast training, little data perceptual needs, and the ability to apply to lightweight, real-time applications. Yet, it is only effective in case of the similarity of domains; mismatched target and source domain may result in negative transfer, and pre-trained layers are frequently correlated with hard-optimization fine-tuning.

IV. ANALYSIS

The paper carries out an in-depth comparison of techniques used in division of medicinal plants of deep learning basics such as individual models, ensembles and transfer learning. For a complete understanding, the chosen metrics were accuracy, training time and resource requirements.

A. Analysis of models in individual method category

The analysis has been done by considering the studies already done on this field, accuracy, training time and complexity was the main key features under this to make a conclusion about the overall model performance of each individual models. Even though many structures offered excellent accuracy, MobileNetV2 was proved to be the most practical and efficient deep learning model for medicinal plant identification. Even though VGG16 (MPInet) achieved the greatest reported accuracy of 99.4%, it took more resources and time to train. Moreover, MobileNetV2 achieved 97.14%

accuracy and had a lightweight design, fast training and was simple to deploy in mobile or embedded setups.

B. Evaluation of Ensemble applied Models

Top results were regularly seen when ensemble models combined the capabilities of several deep learning models. Out of them, EDL-AMLI was especially impressive, hitting 99.66% accuracy in testing and even 99.9% accuracy during cross-validation. It is due to this reason, it is a very reliable and accurate classification of medicinal plants. Using another type of ensemble setup, for instance VGG16 + ResNet50 + ResNet152, produced high accuracy over 99%. Even so, this performance resulted in extra costs: the models were big, had lots of moving parts and it took ages to train them.

C. Evaluation of Transfer Learning Technique:

Experiments compared how effective it is to use pre-trained models for medicinal plant classification, preserving high accuracy with only a little additional training. To observe this performance analysis of some transfer learning model is done in this paper. Transfer learning models were found to give impressive classification results with a lot less effort needed for training. In InceptionV3 + SVM, researchers got the highest accuracy score of 99.81% while using mostly straightforward methods and a moderate amount of training time. It is also a very high performance (99.50% accuracy) and fastest to train, which makes it an appropriate application where performance is required to be very fast and efficient. Very precise but slow to train, and thus the most complex, were architectures like BiT s-r50x1. The Models like AlexNet and VGG16 could train faster, even if their accuracy was not as good if compared to ResNet. All in all, transfer learning methods worked very well, combining high accuracy, good efficiency and the ability to adapt, especially when used with specialized datasets. They provide the right fit for situations where accuracy and processor speed must be balanced such as in scaling settings.

TABLE I. COMPARISON TABLE FOR ANALYSING THE DL MODELS USED FOR MEDICINAL PLANTS

Method Category	Best Performing Model	Performance Metrics		
		Accuracy (%)	Training Time	Resource Requirement
Individual Models	VGG16 (MPInet) [15]	99.4	High	High
	CNN [18]	98.87	Moderate	Moderate
	MobileNetV2 [10]	97.14	Low	Low
	Extended CNN [13]	82.29	Low	Low
Ensemble Methods	EDL-AMLI [39]	99.66	High	Very High
	VGG16+ResNet50+ResNet152 [26]	99.11	Very High	Very High

	HybNet (Feature Fusion) [31]	96.76	High	High
	VGG16+ResNet50+MobileNetV2 [30]	95	Moderate	Moderate
Transfer Learning	InceptionV3 + SVM [34]	99.81	Moderate	Moderate
	BiT _{s-r50x1} [36]	99.62	High	High
	MobileNetV2 (Fine-tuned) [35]	99.5	Low	Low
	VGG16 (Fine-tuned) [33]	96.02	Moderate	Moderate

According to the comparative analysis, the model of InceptionV3 + SVM in transfer learning was the most balanced in terms of accuracy (99.81%), requirements of training, and resources. Although, EDL-AMLI (ensemble) and VGG16 (MPInet) (individual) had comparable results, they require a much larger amount of resources. Compact models such as MobileNetV2 are very efficient when it comes

to making real-time deployments at the cost of fewer accuracy.

V. RESULTS AND DISCUSSIONS

Using individual approaches, ensemble learning and transfer learning demonstrated the most balanced and practical way to classify medicinal plants. Whereas ensemble models, such as EDL-AMLI, were very accurate (up to 99.9 percent with cross-validation), they took a significantly longer time to train as compared to individual models which makes them inappropriate to be used in areas where time and computing resources are scarce. To illustrate, MobileNetV2 could be the fastest to train with 60 seconds and it was good consuming minimal computing power yet earned a few points lower on accuracy (97.14%). The result of training InceptionV3 + SVM combination was just as precise as the use of ensembles, yet it did not use as many resources and was easier. This brought together existing models and simple classifiers which allowed for the desired levels of performance, scalability and feasibility of deployment. So, transfer learning is especially useful for practical purposes, as it combines accuracy with efficiency.

TABLE II. FINAL SUMMARY BASED ON THE ANALYSIS

Technique	Test Accuracy	Training Time	Complexity (1–10)	Strengths	Weaknesses	Overall Verdict
Individual (MobileNetV2)	97.14%	60 sec	3	Fast training, lightweight, suitable for real-time/mobile deployment	Lower accuracy compared to others	Best for real-time, lightweight deployment
Transfer Learning (InceptionV3 + SVM)	99.81%	150 sec	7	High accuracy, efficient feature extraction, scalable	Moderate hybrid complexity	Best overall balance of performance and efficiency
Ensemble (EDL-AMLI)	99.66% (CV: 99.9%)	220 sec	9	Highest accuracy, robust generalization	High training time and computational demand	Best for maximum accuracy when resources allow

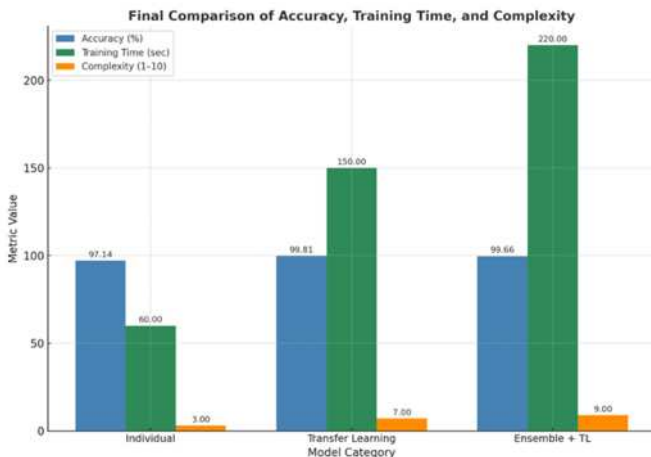


Fig. 1. Table for comparing dl models used for medicinal plants. Final comparison of the top performing models in each category (Individual, Transfer Learning, and Ensemble + TL) evaluated against three criteria: accuracy (%), training time (sec.) and model complexity (1–10 scale). The accuracy of Ensemble + TL is unbeatable, but it costs more time and effort to set up.

It is visible from Fig. 10. that Ensemble + TL based model produces the most accurate results (99.66%) so far, but it also requires the longest time for training (220 sec) and the highest complexity (9). The Transfer Learning model might be the best option if we are looking for a balanced alternative, it has 99.81% accuracy and is less complex (7), plus it takes less time to train.

VI. CONCLUSION AND FUTURE SCOPE

Multiple approaches to deep learning were used in this work to identify plants, giving importance to accuracy, time involved in training and model complexity. It tested individual, transfer learning and ensemble-based models to see which was best for automating medicinal plant identification. Although this paper comparatively reviews the deep learning models according to their accuracy, training time, and their qualitative complexity, it is also noteworthy that, most of the reviewed articles failed to report standardized computing measures like FLOPs (Floating Point Operation), parameters and memory consumption. These quantitative measures are critical in the measure of effectiveness of models and their application in the field.

The involvement of such metrics should be recommended to the authors of future research in order to achieve fair and repeatable comparison. It would also be helpful to consistently provide information about computational cost as well to enable researchers and developers select models not only on basis of its performance but also on basis of hardware availability, energy, and deployment cost. EDL-AMLI that employs an Ensemble + Transfer Learning is the model that recorded 99.66 percent as well as 99.9 percent accuracy during testing and cross validation respectively. The main weakness of all high-performing models is their reduced interpretability and transparency, which is referred to as being in the black box. Due to this, it is worrying in terms of its contribution in fields such as pharmacognosy and study of medical herbs, which demand knowledge as to the reasons that the prediction is made.. This issue reveals a research gap because current models achieve high accuracy, yet do not tell us why they made those predictions. This issue may be resolved by using methods from Explainable AI (XAI) such as Grad-CAM and SHAP which identify the aspects of an image that greatly influence a model's prediction.

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